

Sharoon

Software Engineer

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SUMMARY

Software engineer with a strong backend focus. Build production systems end-to-end — REST APIs, event-driven services, data pipelines, and cloud infrastructure on Node.js, NestJS, MongoDB, AWS, and GCP, with frontend work in Angular and React. Experienced across architecture, implementation, optimization, deployment, and production support. Track record of owning performance, scalability, and reliability — eliminating API crashes under high-volume load, cutting multi-second queries to sub-second, and shipping AI-driven systems, serverless orchestrations, large-scale data migrations, and internal developer tooling that measurably improves team velocity.

EXPERIENCE

Software Engineer — Recotap

Present

Product company building an ABM marketing platform on LinkedIn. Products: Recotap and AdRadar.

- Built two AI auto-blocking agents (company and title classification) integrated into the campaign delivery flow, removing manual block-list curation and improving targeting precision.
- Designed serverless workflows on AWS Step Functions and Lambda to replace fragile cron pipelines — multi-stage jobs with retries, parallel branches, durable state, and dead-letter queues.
- Owned the GCP Pub/Sub internal notification system with topic-per-event design, idempotent handlers, and a thin SDK for easy adoption across services.
- Owned the LinkedIn Marketing API integration end-to-end — rate-aware sync, data normalization, and tuned MongoDB aggregations that keep dashboard P95 under 200ms.
- Owned performance, scalability, and reliability: eliminated API crashes under ~2,000 req/sec CRM sync load by offloading ingestion to a BullMQ + Redis queue (HTTP 202 in ms), and cut a Sales Activity API over ~7.5M records from 50s+ (504 timeouts) to 2–3s (~20x faster).
- Built a shared Redis connection pool (bounded 10–500) to end connection leaks and exhaustion fleet-wide, and fixed a pagination data-integrity bug via a deterministic sort tiebreaker and idempotent upserts.
- Built Angular frontend surfaces for the ROI analytics and integration features delivered on the backend; operated services at 99.9% uptime.

Software Engineer — Skills Outsource Think Pvt Ltd

Earlier

Service company running ad operations for French advertisers across Meta, TikTok, and other social platforms. Product: Haiku.

- Led a complete production data migration from Airtable to MongoDB — schema design, batched ETL with checkpointing, dual-write cutover, and full row-level validation with 100% data parity.
- Designed and shipped an internal MongoDB query-builder plugin with a typed, composable API — reduced team-wide query-writing effort by approximately 40%.
- Built a seamless cross-domain auto-login (SSO) flow with short-lived signed tokens, CSRF-protected handshake, and silent re-authentication.
- Built REST APIs and frontend features in React across customer-facing and internal products; mentored teammates on MongoDB internals and API design via code reviews.

PROJECTS

AI Auto-Blocking Agents (Company and Title)

Recotap and AdRadar

- Designed an LLM-backed classification pipeline that scores candidate entities with auditable reasoning, with confidence-threshold routing for human review.
- Integrated decisions directly into the campaign delivery flow so blocks take effect in the next ad-serving cycle.

Tech: Node.js, NestJS, TypeScript, MongoDB, LLM APIs, AWS Lambda

State Machine and Lambda Orchestration

Recotap and AdRadar

- Modeled long-running multi-stage jobs as AWS Step Function state machines with explicit retry and failure transitions.
- Broke heavy work into parallel Lambda tasks with dead-letter queues and CloudWatch alarms; made state durable so jobs survive timeouts and resume from the last completed step.

Tech: AWS Lambda, Step Functions, CloudWatch, Node.js, TypeScript

GCP Pub/Sub Notification System

Recotap and AdRadar

- Designed a topic-per-event model so producers publish without coupling to consumers.
- Built idempotent handlers, dead-letter routing, and structured retries, plus a thin internal SDK so any service can publish or subscribe in a few lines of code.

Tech: GCP Pub/Sub, Node.js, NestJS, TypeScript

LinkedIn Integration and ROI Analytics

Recotap and AdRadar

- Built a rate-aware LinkedIn Marketing API sync that respects quota and resumes cleanly from interruption, normalizing campaign, spend, and outcome data into a query-friendly schema.
- Designed aggregations and indexes tuned to the ROI dashboard access pattern, keeping P95 under 200ms; built the Angular ROI graph.

Tech: LinkedIn Marketing API, MongoDB, Aggregation Pipeline, Angular, Node.js

High-Volume CRM Sync Offload (BullMQ Queue)

Recotap and AdRadar

- Diagnosed API crashes from a synchronous CRM sales-activity endpoint hit at ~2,000 req/sec (~500 records each), exhausting the tenant DB connection pool and spiking MongoDB CPU.
- Re-architected it to enqueue a job and return HTTP 202 in milliseconds via a dedicated BullMQ + Redis queue (bounded concurrency, backoff, retries) — database load now stays flat regardless of inbound spikes.

Tech: BullMQ, Redis, Node.js, NestJS, MongoDB, TypeScript

Sales Activity API Optimization (50s+ to 2–3s)

Recotap and AdRadar

- A search, pagination, and total-count list API over ~7.5M records took 50s+ and routinely threw 504 Gateway Timeouts on large tenants.
- Replaced unindexed \$regex full-collection scans with index-backed cached lookups, redesigned compound indexes with query hints and maxTimeMS guards, and parallelized count/data queries with short-TTL caching — cut latency ~20x, eliminated 504s, and sharply reduced MongoDB CPU.

Tech: MongoDB, Indexing, Aggregation Pipeline, Redis, Node.js, NestJS

Airtable to MongoDB Migration

Haiku

- Mapped Airtable bases to a MongoDB schema designed around production query patterns; built a batched ETL with checkpointing and a dual-write phase to validate before cutover.
- Wrote validation scripts comparing row counts and field hashes between source and target (100% data parity).

Tech: MongoDB, Airtable API, Node.js, TypeScript, Batched ETL

MongoDB Query-Builder Plugin

Haiku

- Extracted common filter, projection, and aggregation patterns into a typed, composable API where queries are built declaratively and verified at compile time.
- Rolled it out gradually with adoption docs and migration examples — reduced query-writing effort by approximately 40%.

Tech: MongoDB, TypeScript, Node.js

Cross-Domain Auto-Login (SSO)

Haiku

- Issued short-lived signed tokens that one domain can exchange for a full session on another.
- Built the cross-domain handshake with CSRF protection and strict origin validation, plus silent re-authentication for transparent session refresh with graceful fallback to native login.

Tech: Node.js, JWT, Secure Cookies, TypeScript

SKILLS

Languages: TypeScript, JavaScript, SQL

Backend: Node.js, NestJS, Express.js, REST APIs, API Gateway, Microservices

Frontend: Angular, React, HTML, CSS, Bootstrap

Databases: MongoDB, Aggregation Pipeline, Indexing, Query & CPU Optimization, Bulk Operations, Redis, Airtable

Messaging and Queues: BullMQ, Kafka, RabbitMQ, GCP Pub/Sub

Cloud and Serverless: AWS Lambda, AWS Step Functions, GCP, Vercel

DevOps: Docker, Kubernetes, CI/CD Pipelines, Production Debugging

Architecture: Event-Driven Design, State Machines, Performance & Reliability Tuning, API Optimization, Connection Pooling, SSO and Authentication

Engineering Leadership: Code Review, Mentoring, Technical Specifications, Architecture Reviews, Observability, Cross-Team Collaboration